Suggested problems: 1-37 odd

Problem 1: The temperature adjusted for wind chill is a temperature which tells you how cold it feels, as a result of the combination of wind and temperature; see the table below.

		Temperature (°F)							
		35	30	25	20	15	10	5	0
Wind Speed (mph)	5	31	25	19	13	7	1	−5	−11
	10	27	21	15	9	3	−4	−10	−16
	15	25	19	13	6	0	−7	−13	−19
	20	24	17	11	4	−2	−9	−15	−22
	25	23	16	9	3	−4	−11	−17	−24

(a) Express the dependency of wind chill on wind and temperature in function notation.

(b) If the temperature is 0°F and the wind speed is 15 mph, how cold does it feel?

(c) If the temperature is 35°F, what wind speed makes it feel like 24°F?

Problem 2: A car rental company charges \$40 per day and 15 cents a mile for its cars.

(a) Write a formula for the cost, C, of renting a car as a function, h, of the number of days, d, and the number of miles driven, m.

(b) Find $h(5, 300)$ and interpret it.

Problem 3: Suppose $f(x, y) = x^2y + \sqrt{x}$.

(a) $f(1, 2)$

(b) $f(9, 4)$

Problem 4: Consider the three points

$$\begin{aligned} A &= (1.3, -2.7, 0), \\ B &= (0.9, 0, 3.2), \\ C &= (2.5, 0.1, -0.3), \end{aligned}$$

in $\mathbb{R}^3$.

(a) Which one is closest to the yz-plane?

(b) Which one lies on the xy-plane?

(c) Which one is farthest from the xz-plane?